Matematical Analysis

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Contents

1 Find the particular solution satisfying the initial condition for the given differential equation

Given the differential equation with initial condition $y(1) = \frac{\pi}{2}$:

$$xy' - y = x \tan \frac{y}{x}$$

1.1 Rewrite and Identify Type

Dividing both sides by x to isolate y':

$$y' = \frac{y}{x} + \tan \frac{y}{x}$$

The right-hand side depends only on $\frac{y}{x}$, so this is a homogeneous equation.

1.2 Substitution

We introduce the substitution $z = \frac{y}{x}$, i.e. $y = zx$, so that:

$$y' = z'x + z$$

Substituting into the equation:

$$z'x + z = z + \tan z$$

$$\frac{dz}{dx}x = \tan z$$

1.3 Separate Variables and Integrate

Separating variables:

$$\frac{dz}{\tan z} = \frac{dx}{x}$$

Integrating both sides:

$$\int \frac{dz}{\tan z} = \int \frac{dx}{x}$$

The left integral simplifies by rewriting $\frac{1}{\tan z} = \frac{\cos z}{\sin z}$:

$$\int \frac{dz}{\tan z} = \int \frac{\cos z}{\sin z} dz = \int \frac{d \sin z}{\sin z} = \ln|\sin z| + C$$

And the right integral:

$$\int \frac{dx}{x} = \ln|x| + C$$

Equating both results and exponentiating:

$$\ln|\sin z| = \ln|x| + C$$

$$\sin z = x \cdot C$$

1.4 Apply Initial Condition

Substituting back $z = \frac{y}{x}$:

$$\sin\frac{y}{x} = x \cdot C$$

Applying the initial condition $y(1) = \frac{\pi}{2}$:

$$\sin\frac{\pi}{2} = C \Rightarrow C = 1$$

Therefore the particular solution is:

$$\sin\frac{y}{x} = x \Rightarrow \frac{y}{x} = \arcsin x \Rightarrow \boxed{y = x \arcsin x}$$

2 Find the general solution for the given differential equation

Given the differential equation:

$$y'\left(x+y^2\right)=y$$

2.1 Rewrite as a Linear ODE

Rewriting the equation with $\frac{dy}{dx}$ on the left:

$$\frac{dy}{dx}=\frac{y}{x+y^2}$$

Taking the reciprocal to treat x as a function of y instead:

$$\frac{dx}{dy}=\frac{x+y^2}{y}$$

$$\frac{dx}{dy}=\frac{x}{y}+y$$

Rearranging into standard linear form:

$$\frac{dx}{dy}-\frac{x}{y}=y$$

2.2 Solve the Homogeneous Equation

We first solve the associated homogeneous equation $\frac{dx}{dy}-\frac{x}{y}=0$:

$$\frac{dx}{x}=\frac{dy}{y}$$

$$\ln|x|=\ln|y|+C$$

$$x=y\cdot C$$

2.3 Variation of Parameters

We allow C to be a function of y, setting $x=y\cdot C(y)$ and differentiating:

$$x'=C(y)+y\cdot C(y)'$$

Substituting into the original equation:

$$C(y) + y \cdot C(y)' - \frac{\cancel{y} \cdot C(y)}{\cancel{y}} = y$$

$$\cancel{C(y)} + y \cdot C(y)' - \cancel{C(y)} = y$$

$$y \cdot C(y)' = y$$

$$C(y)' = 1$$

Integrating to find $C(y)$:

$$C(y) = \int dy = y + C$$

2.4 General Solution

Substituting $C(y) = y + C$ back into $x = y \cdot C(y)$:

$$\boxed{x = y^2 + y \cdot C}$$